

Mathematics A

Unit 2 • Chapter OL-T30 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

7 classified items • 19 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T30 • Expertise Q16+ • 7 items • 19 marks

Chapter	Items	Marks	Starts
01. OL-T30 Functions	7	19	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Functions

Unit 2 • Expertise • 19 marks • 7 item(s)

June 2025 | 4MA1-1H Q16 | 1 marks

Evaluate a function at a numerical input

June 2025 | 4MA1-1H Q16 | 3 marks

Solve a composite function inequality

June 2023 | 4MA1-1H Q17 | 4 marks

Exclude a denominator zero from the domain

January 2022 | 4MA1-2H Q25 | 1 marks

Domain and range swap under inversion

June 2021 | 4MA1-1H Q24 | 2 marks

Evaluate a numerical composite function

June 2021 | 4MA1-1H Q24 | 4 marks

Inverse of a restricted quadratic function

January 2023 | 1H Q18 | 4 marks

Inverse of a composite-defined function

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$$g(x) = 4x^2$$

(a) Find $f(9)$

(1)

WORKING SPACE

$$g(x) = 4x^2$$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

WORKING SPACE

4MA1-1H Q17

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....
(1)

(b) Solve $gh(x) = 1$

.....
(3)

WORKING SPACE

4MA1-2H Q25 Q25(b)

1 marks

(b) State the domain of g^{-1}

(1)

WORKING SPACE

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

YOUR WORKING

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

YOUR WORKING

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

YOUR WORKING
